Rehabilitation of Distribution Networks for Reduction in Line Losses and Power Quality Improvement

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To my loving and caring family

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Abstract:

Electrical energy is basic human need and increasing day by day with ever increasing advancements in the industry of Electrical Auto Mobile industry and electrical household uses. In every third world country there is a short in demand of electricity. The power generated always is not enough for the power distributed among and with ever increasing load value. AS in case of Pakistan, the demand of electricity in Pakistan is 26000 MW, And the amount of power generated is 22000 MW, and among them 4000 are always lost in line losses. Therefore, it is not enough to generate the high amount of power. But also, there is need for efficient transmission as well. The main cause of power losses subjected to line of the system is line losses, harmonics distortion which results in increasing the tremendous amount of power losses. Therefore, this thesis makes a concern on decreasing the reasonable amount of power losses as in form of power losses and improvement in voltage profile of the system using two of the practice methods OCP and DG. Optimal capacitor placement and Distributed Generation are the two methods by placing them at position we can take care of line losses of the system. A comprehensive 11 KV feeder from Sahiwal grid consisting of 177 buses based on actual comprehensive data using ETAP and run the load flow of the required system before and after placing the OCP and DG respectively. Line losses of the system were comprehensively calculated before and after the solution was implemented. The position of Optimum Capacitor Size is defined by placement formula and placement and size of DG is performed by Genetic Algorithm. Then we need to examine the power loss and voltage by three stages for example

- With only OCP
- With only DG
- With both OCP and DG.

The type of DG was also changed accordingly for change in working condition in environment. we used

- One wind and one solar
- Both solar

And then the results are examined after placing both units one by one. Graphs are drawn after the placement of both OCP and DG accordingly.

Chapter 01. Introduction:

In an electricity dependent world, we the human beings, are so dependent on electric charge that it has become an essential part in sustaining the modern ages of life. Electricity requirements have been drastically boosted up in the recent years due to advancements in the industrial and scientific pattern of life. Now, with the increase in demand, the handling parameters of such a big amount of electric charge have also been taken into the consideration. Because with the increase in the size of a system, there is an obvious decrease in the efficiency. Now additional measures must be taken to achieve the required efficiency to meet the economic needs.

Modern grid systems are an example of an advanced system of electricity production and safe transmission at a larger scale. NETL a U.S based initiative is doing massive work in this field. They are working on the reliability, security, cost of service, power quality, efficiency, environmental impact and safety of a power grid system. Moreover, there are different kinds of modern grid systems like radial and interconnected grid systems. Each is used according to the map and consumption tactics of our distribution network.

The distribution networks lose their efficiency because of the line losses and voltage dips along their transmission. Because when electricity is transmitted at long distances, there occur voltage drops, and the transmission of reactive power is affected. So, the source generators are distributed along the network in such a way to make the distance between the source and consumption end, minimum. And in this way, we achieve a major portion of our electricity transmission requirements.

1.1 Energy requirements:

Since Pakistan is a developing state so the electricity sector of Pakistan is also advancing day by day. The economy of Pakistan in being affected by the risky and uncertain electricity distribution networks which are not reliable. WAPDA which holds power over transmission, generation and distribution to energy consumers, except the city of Karachi locates in the province of Sindh where K-electric is responsible for power distribution, was sub-dived for the various regions into 10 distribution companies, 4 thermal power generation companies which are owned by the government

and a company responsible for transmission having the name National Transmission and Dispatch Company (NTDC).

Pakistan has a total installed power generation capacity of over 40,000 MW as of 5 April 2021. Furnace oil (15 percent), hydel (26 percent), natural gas (12 percent), LN G (25 percent), coal (9 percent), renewable (solar and wind 5 percent) and nuclear (8 per cent) are the principal sources.

When the long paths of transmission are there in the distribution networks, the efficiency is reduced due to the voltage drops across the path. To meet the drops of voltage across the path of a distribution network, there is efficient way to use. Distribution generators are used across the transmission at points where there is a drop of voltage and these distribution generators are some sort of renewable energy resources like solar, thermal, wind etc. Moreover, different transmission methods are being utilized to meet the efficient transmission requirements.

Microgrids are another solution to the energy distribution problems in Pakistan prese nt a viable alternative to the centralized grid infrastructure. Microgrid enhances energ y resilience and recovery, reduces energy costs for customers and businesses as energ y is now produced near to the consumer that helps reduce line losses and theft.

1.2 Objectives:

Electricity has become the important part of modern life. We use electricity for lighting, heating, cooling, and refrigeration and equipment, computers, appliances, machinery, and public transportation. Demand for electricity is growing day by day. So, to meet with the increasing demand the distribution networks are extended day by day on regular basis. However, this extension if done without proper planning, may increase energy losses and cause power quality problems. This results in revenue loss to the utility and poor services to the consumers so, reduction in line losses and power quality improvement has become the major concern.

To address this situation various corrective measures are taken including installation of capacitors, reconductoring, feeder bifurcation, and transformer relocation and upgradation. To get full benefit out of this rehabilitation process, optimal selection, sizing, and location of the equipment is very important. The research will focus on rehabilitation of a DN by determining optimal size, location and number of capacitor

banks, transformers and distributed(renewable) generation by considering benefit to cost ratio. A strategy will be devised for reduction of energy losses and improvement in power quality that is applicable to DNs in Smart Grid context.

1.3 Approach and Methodology:

Software-assisted load flow analysis is a great technique to get accurate and dependable findings. In this case, we used the Electrical Transient Analyzer Program (ETAP) to perform load flow study on an 11kV feeder Distribution Transformer ratings and values, as well as transmission transformer ratings and values

ETAP was used to have a brief analysis of the wires and loads that were collected over several days.

Causes of these losses and Methods mostly used for this process are given below.

1.3.1 Why there are Line losses??

Electricity is generated from the generating end and stations, while distribution of the power has some important role of itself as well in it. The power quality of the power system is measured by the voltage profile and the regularity and reliability. As we all know Load at the distribution end is mostly inductive and having low PF profile. Mostly we need a large amount of reverse reactive power to compensate the inductive effect of the load as well. This as well has some effect on the Voltage dip came across the system. As these voltage dips are the cause of the power losses in the system. For this we must compensate this effect as well. Third issue is $Losses = I^2R$ When we look at the resistive effect of the wire, we are working along with we got this as a major effect. when we talk about the Mostly this can be done by few methods given below

- Capacitor Banks.
- Synchronous Condensers.
- Phase Advancer
- Distributed Generator

We are talking about only two of them.

1.3.2 Optimum Capacitor Banks:

We goanna present the proposes two-stage techniques for determining the size and location of a capacitor bank.

The load flow is carried out in the first step to find the losses of the system. The sweep algorithm is used by the system. Different sizes will be used in the future stage.

Each contender has a set of capacitors that have been initialized and placed in it. The system's load flow is carried out by bus and then again by the system's load flow. The goal is to carry out the process with minimum number of losses The cost function, which includes the cost of the capacitor and the cost of the loss, Voltage limitations were used to confine the formulation. The capacitor that contains the

The solution with the lowest cost is chosen as the best option. The suggested approach is used on a variety of conventional test systems.

The value of a capacitor is determined using the methods described based on the cost of active power losses and the cost of installing a capacitor bank in the distribution n etwork. The procedure begins with a standard load flow procedure. Following the loa d flow procedure, the capacitor size (QVAR) is randomly initialized between zero an d two times the total reactive load of the system Qload i.e., the lowest value is define d as zero and the highest value is defined as two times the total reactive load of the s ystem Qload.

1.3.3 Advantages of OCB:

The only to transmit power from one point to other is only from bus-to-bus system. There is no direct method to transmit power. We can’t even track the original bus from any of the system by just analyzing it. The optimal placement of capacitor is generally a hard combinatorial optimization problem that can be formulated as a nonlinear multi-objective problem. A way to reduce the chances of overcrowding in a local power distribution network has been achieved by removing congestion and using a local energy storage device. We can analyze the system by using optimum analysis through ETAP. The basic advantage we take is the Improvement of power Quality**,** reducing the Inductive effect through Capacitive**,** improving Voltage Regulation, improvement of the Cost losses**,** Compensation HT load requirement.

1.3.4 Capacitor Installation cost:

The proper cost analysis for a feasible solution can only be done by the ratio of both cost of the power being lost and the cost of the system to be installed. The power los s of the branch is calculated by line current I and branch resistance R as P= I2 R. Tot

al power loss is checked by adding the whole line. The power loss provided by the lo ad flow process is the maximum power loss. Medium power loss is calculated using the load factor loss, The cost of a capacitor depends on the size of the capacitor bank to be installed. Although the cost of the size of each capacitor unit varies from size t o size, that is, per unit cost of lowcapacity capacitor banks is very similar to the unit cost of a large capacitor bank. The life of an installed capacitor bank is considered to be 10 years.

The DG is an electrical energy source that links the distribution system. Because most of the DGs are renewable energy sources like PV models and Wind Turbines (WT), it creates clean energy. Power losses are reduced, efficiency is increased, power quality is improved, system dependability is improved, and fuel costs are reduced, i.e., cost of maintenance and operation. However, if the DG is installed in the incorrect place or has insufficient capacity, it will result in more power loss and expense than if it is not installed at all. Typically, distributed production means the production of electricity within a distributed network to meet the immediate energy demand of consumers. The DG can be treated as a small power producer to reduce the consumer energy demand. Distributed Generation can come from a variety of sources and technologies. There are many similar sources of production. The actual power loss of the distribution system depends on the resistance of the distribution lines, the main losses of transformers and motors. Since the dielectric and rotational losses are very small compared to line losses, therefore, only line losses are considered in this analysis.

1.3.5 Advantages of DG:

Reduced natural emissions through spotless and sustainable innovations (Green Power), providing cresting capacity to lower the cost of power, CHP, a higher level of unwavering quality and nature of the power that is being supplied, and deferral of the transmission and distribution line venture through improved load ability are just a few of the significant applications of the DG. Reduced natural emissions through spotless and sustainable innovations (Green Power), providing cresting capacity to lower the cost of power, CHP, a higher level of unwavering quality and nature of the power that is being supplied, and deferral of the transmission and distribution line venture through improved load ability are just a few of the significant applications of the DG.

The size and position of the DG are quite important., As the DG size is raised, the losses reduce, but once we pass the optimal DG size value, the losses begin to climb again,

and in certain circumstances, they may overshoot from the base case. Location is also considered, since an optimal DG size on one bus may not work well on another, resulting in increased losses. Another important factor to remember is that no matter what, a large enough DG size to export electricity out of the substation will never be chosen, which will result in larger losses.

1.3.6 Cost Efficiency of OCB & DG:

The objective function is to analyze the two factors. Both is the:

Power Loss cost

Capacitor Installation cost

The proper cost analysis for a feasible solution can only be done by the ratio of both cost of the power being lost and the cost of the system to be installed. The power loss of the branch is calculated by line current I and branch resistance R as P= I2 R. Total power loss is checked by adding the whole line. The power loss provided by the load flow process is the maximum power loss. Medium power loss is calculated using the load factor loss, The cost of a capacitor depends on the size of the capacitor bank to be installed. Although the cost of the size of each capacitor unit varies from size to size, that is, per unit cost of low-capacity capacitor banks is very similar to the unit cost of a large capacitor bank. The life of an installed capacitor bank is considered to be 10 years.

Chapter 02: Literature Review:

2.1 Significance of Rehabilitation:

The distribution network is scheduled for a limited time (5 years) and their network expanded; its old (existing) part should also be appropriately added. However, to meet the growing expanded without adding the existing network approp riately, hence the loss of power and power outages are increased beyond the permissi ble limits. Engineering economics and finance both show their declining existence as well future extensions. To reduce power loss and voltage reduction alternatives are a vailable approved to expand and revitalize the existing distribution system. The main objectives of rehabilitation are given.[1]

- Reduction of power losses
- Improvement of voltage conditions
- Improvement of power factor
- Improvement in safety
- Improvement of customer service
- Reduction in cost of operation and maintenance
- Improvement of equipment life

The distribution network is selected for renewal based on specific planning condition s such as power outage, power loss, equipment loading, profit / cost ratio etc. For eac h selected network, a Power Loss Reduction planning proposal is prepared to achieve the benefits you want. There are the following factors which cause the line losses an d voltage drop in distribution system.[1]

- Impedance of line conductor
- Loading of line conductor
- Power factor
- Poor maintenance
- Theft of electricity

There are two types of special losses

2.1.1 Fixed Technical Losses:

The fixed loss does not vary according to the current. These losses are in the form of heat and noise and continue to occur as long as the transformer is in a state of power, between one-fourth and one-third of the technological losses in the distribution network is coming under the fixed loss category. Consistent losses in the distribution network may be affected; Current Leakage Loss, Corona Loss, Dielectric Loss, Open Circuit Loss, Loss due to the continuous load of various measuring objects, and Loss caused by the continuous load of various control elements. [3]

2.1.2 Variable Losses:

The variable losses vary in the size of the transmitted power as well, depending on the current square. Therefore, the current 1 percent increase encourages an increase of a maximum of 1 percent loss. Between two-thirds and three-quarters of the technical losses in distribution systems fall into the category of variable losses. By increasing the cross-sectional area of the transmission lines and the specific load links, the losses will be reduced significantly. This nonetheless brings an exchange between the cost of loss and the cost of capital expenditure. It is recommended that the average usage rate in a supply system that considers the cost of loss in its system can be as little as 30 percent. Examples are Impedance losses, losses brought by touch opponents, and Joule losses present on conductors at all power levels. [3]

2.1.3 Main System contributing losses:

Each part of the service transfer once the distribution system contributes to the loss, so losses are avoided at the end of the customer use or meter combinations, moving support the system at production level. Table below shows the typical line loss in each section below reception

Loss of transmission system line usually involves two (or more) changes sections an d one (or more) additional set of lines. Depending upon voltage and distance, loss of transmission line from two to five percent.

[4]

Losses at Each Stage of Electricity Distribution[6]

Component	Estimated Loss as a Percentage of Energy Sold	
	Typical Urban	Typical Rural
Subtransmission Lines	0.1	0.7
Power Transformers	0.1	0.7
Distribution Lines	0.9	2.5
Distribution Transformers No Load	1.2	1.7
Distribution Transformers Load	0.8	0.8
Secondary Lines	0.5	0.9
Total	**3.6**	**7.3**

The following section describes each section of transmission and distribution system, with reference to how losses occur and how they can be reduced.

2.1.3.1 Step-Up Transformers.

These are transformers found at generating facilities. They convert voltages that are s uitable for transmission line. Large standard generators generate power at 6600 volts, 13,800 volts, 18,000 volts, or even 22,000 volts, while normal transmission voltages in United States 115 kV, 138 kV, 230 kV, 345 kV, 500 kV, and 765 kV. Step-up transformers are generally measured in productive units, with minimal losses to n ormal operating levels. If, however, they have more power than their first targeted ca pacity, losses increase significantly. This can be a problem when productive units are "upgraded" to provide higher output without the same development of step-up transformers. Also, if generators operate with an unbalanced power factor (describ ed below), causing an increase in operating power output (also described below) may increase system losses at all levels.[5]

2.1.3.2 Transmission system conductor

Long distance transmission lines bring power from the generator to the distribution s ervice area. Although the conductors themselves have low resistance, the length of th e lines and the magnitude of the affected loss of the conductor. Line losses can be sig nificantly reduced by current long-distance transmission (DC) systems, making DC transmission desirable for long-distance transmission lines. However, the additional losses are up to 1.5 percent from the conversion stations at each end of the DC transmission line.[5]

2.1.3.3 Distributing Stations

Many utilities have a step in the middle of their plants, with the power taken from th em "Distribution stations," which receive high voltage (230 kV and above) and bring that power to a constant small distribution station at 69 kV or 115 kV. Transformer l osses that occur in small channels occur twice – first converting power from high-voltage transmission at medium voltage, then again at smaller stations converting it t o the initial voltage. Principal losses in distribution stations transformer losses. The r eason is that resources use different levels of electricity to differentiate the transfer of excess energy from the loadsupplying power. This method increases the reliability of the system. [5]

2.1.3.4 Substation Transformers.

These take power from transmission system, usually 115 kV or more, too converted t o 4 kV to 34 kV transmission lines. They are specially designed for their expected lo ads, themselves it is not uncommon to carry power near that limit to allow load trans fer between circuits, but there are two problems of concern. First the major losses are likely to be very high when loaded slightly. Second is the contradictory loss which may be too high if loaded too much.[6]

2.1.3.5 Voltage Regulators.

These are four transformers many taps installed near distribution circuits in order allo w to increase or decrease the variable voltage points. Historically these were installed in long rural areas distribution lines to enable remote power outages points, removin g the reduced voltage caused by line resistance. Today there are more jobs for these devices. In addition, they enable strong conservation power control in the distribution line, which may reduce total power consumption. Because they are transformers, inc luding both core losses as well loss of resistance, and attention to both building mater ials as well the magnitude of this affects the rate of loss of lines.

2.1.3.6 Primary Distribution Lines. The main lines connect sub-channels in circuits that bring energy to the business regions and neighboring areas. These usually run at 4 V to 34 kV. High voltage, low current, and thus reducing the opposing losses on these lines. [6] However, high voltages require long (or very expensive poles) underground technology), so there is a cost / efficiency trade.

2.1.3.7 Line Transformers.

These are trash sizes cylinders you see mounted on neighboring electric poles or in m etal boxes mounted on concrete pads. They are changing the first electrical power to distribute the voltage we use in our homes and businesses, usually 120 V, 208 V, 240 V, 277 V, and 480 V. [4]

2.1.3.8 Secondary Distribution Lines.

These connect line transformers to individual homes and businesses. That is usually very short, in part because of this low voltage, the amperage required to transmit pow er is significant, which requires large (and thus very expensive) conductors. Losses c an be very high due to high current. This is especially true if the second load has gro wn faster or faster than the real guess.[4]

2.2 Methods to reduce line losses and Voltage drops:

2.2.1 Reducing Transformer Losses:

Remember that transformer losses are caused by two different mechanisms, primary l oss (noload) and opposing loss (copper). The main loss is the loss that is gained in or der to be empowered transformer. These vary in size of the transformer and materials used to build the transformer. It is important for "correct size" transformers to mini mize major losses. In the case where, for example, a large an industrial customer wit h heavy equipment and highpower demand comes from the main building and is repl aced the work of a warehouse with only lights and a few offices equipment, and no c orresponding conversion is performed on transformers, the main losses can last more than a year power consumption of a new business. Opposition losses are primarily a f unction of the present flowing through the transformer, burning it. [21] This is a loss they explain the current ones. For this reason, it is important not to have a very small transformer, or it will "burn" at a great loss. Another option is for resources to inclu de banks of three or more transformers in small, exhausting stations one or more tim es of low load (to avoid overweight loss), but then opens them at the most demandin g moment times (to avoid strong loss of opposition). And, it is possible it has been a trade-off resulting in increased circuit breaker costs and lower risk of fidelity.

2.2.2 Reducing Line Losses:

All standard quality conductors are made very pure aluminum or copper, both natural ly low electrical resistance. There are three aspects to that with a significant impact o n the loss of conductors. [8] The first is quality of communication at each end of the conductors (and any possible episodes in the middle of the line). The second I the siz e of the conductor relative to the amperage it is carrying. I the third is the voltage at which the conductors operate. Communication quality is generally very good in the United States, but it is the source of the most important line losses for the least devel oped countries. Rusty, or simple connectors twisted cables, resulting in significant el ectrical wiring current. The size of the conductor affects the resistance of the line to c urrent passing through it. When high amperage is expected, large conductors are req uired, such as a large gauge an extension cord is required to carry electrical tools and so on widely used electrical appliances of heat. Resources sometimes change cables or "reconductor" is an existing distribution circuit (without changing its voltage) to in crease volume and reduction of losses in that cycle. This is expensive, but it is not as costly as the full reconstruction required increase the voltage. And sometimes there is no other way, as when a single-family residence gradually changes multi-family or commercial development. The voltage affects the loss by reducing the requ ired amperage to deliver any given number of watts to customers. By increasing the voltage in a line - usually that means new transformers should also be installed - utility can lower the line temperature. High power lines and usually require longer p oles, however, as well as costs. Involved in the installation of new poles may prevent . Usage of the underground cable for high voltage power lines are numerous times m ore expensive than top construction and so on it is usually limited to a short distance and in comparison, flat surface. To promote the use of a dispersed generation such as Solar photovoltaic and wind can also greatly reduce the system losses if carefully pl anned.[8] Distributed generation helps with to provide a source of energy near the re ceiving loads of consumption, thus avoiding the need for energy to be delivery to re mote central power stations, suffering getting lost along the way.[8]

2.2.3 Power Factor and Reactive power:

Have a great impact on increasing the electric grid capacity and in reducing the line l osses. A "Power factor" is the number that basically indicates how well a device uses electricity. It is rated as a measure of "real power per kW" to "apparent power per ki lo voltampere (kVA)" in the distribution circuit or end consumption. The difference b

etween the two shows how real power is used in the right way. “Real power” is the e lectrical component that performs useful functions. "Active power" establishes the m agnetic field required by motors and transformers to operate but does not contribute t o useful work. Real power is only generated from generators - as well as a generation that is distributed as solar photovoltaics. Active power can be generated by both generators and capacitors. For maximum efficiency, the generator must operate on its limited power factor or higher.[8] The same is true of motors and other end-of-life items. Blocking loads (such as incandescent light bulbs) have a power rating of 1 .00, which means they use only real power; therefore, the real power and the visible power are like such loads. However, motors, transformers, electronic equipment, and distribution lines consume real and effective energy. Therefore, their power factor is l ess than 1.00 unless the power factor adjustment technology is used. In fact, some en gines (such as those in refrigerators and especially old air-conditioners) and electrical appliances (such as those on personal computers, office e quipment, and televisions) overload the electrical system in excess of the amount of e nergy they use. Productively. While kilowatt hours (kWh) measure the amount of po wer used by the end user, kilovolt-ampere hours measure the total amount of power that should be supplied to the servic e. Modern measurements can point to these differences and can help empower consu mers or resources to take corrective action. This usually involves installing capacitors to provide active power to the customer's equipment instead of requiring a grid to pr ovide all the required operating power.[8] While services often charge large customer s partially for their high level of demand, which includes additional losses due to the negative energy aspect, most small businesses and residential buyers are not charged a much higher price. The main reason for this was that the required measuring equip ment was historically expensive, and residential buyers had fewer loads that created s ignificant energy factor problems. Today both aspects have changed. Modern, inexpe nsive, smart meters can easily measure kilo voltampere hours as they measure kWh, so utilities can charge customers the actual power they need (kVA), not just the ener gy they use (kW). This provides real encouragement for consumers to invest in energ y efficiency. This is no small feat. One of the bestselling home refrigerators, the Whi rlpool 22-cubic foot bottom freezer model, is rated to have a power factor of less than 40 percent, me

aning the kVA capacity required to provide 2.5 times the kW unit used by the unit. T his drive current is currently in the home circuit, second distribution line, line transfo rmer, and so on at the top of the distribution circuit when capacitors are not installed somewhere in the circuit to correct and correct this power factor problem. Because c onductors, transformers, and generators of electricity are measured in kVA and not k W, if this power factor is not adjusted, it increases the cost of the entire electrical sys tem. Also, if left untreated, the high amperage effect placed on the lines and transfor mers also further counteracts loss.[8] Utilities - as well as taxpayers -
will have to spend extra money immediately to replace unnecessary grid equipment. Circuit and channel development and even generation can be reduced or reversed if t he power factor is adjusted. As residential loads shift from opposing loads (e.g., inca ndescent lamps, electric cables, dryers, and electric water heaters) to other electronic and automotive loads (e.g., compressor air conditioning), the resilience feature has be come a promising source. of essentials. reduction of capacity, which makes adjustme nt of the power factor very important in improving the efficiency of the system. The adjustment of the power factor is most effective when done close to the loads involve d, so that the high current does not interfere with the cables rising the river from the l ast use. Federal standards appliance may require a highpower factor and energy effici ency of kWh, but until this is available, and inventory is improved, resources can gai n significant energy benefits and reduce line losses by dealing with commercial and r esidential power factor problems with careful capacitor installation in distribution cir cuits.[8]

2.3 Optimal Capacitors Placement:

The OCP searches optimum capacitor bank locations and sizes for inclusion in the radial distribution server in order to maximize residual economic savings through reduced power losses and power losses, as well as lower investment costs and capacitor bank installation. To be a feasible solution, the OCP solution must overcome physical, engineering, and operational hurdles; as a reason, the OCP is classified as a limited development challenge. Due to equilibrium restrictions, the network has to be able to provide efficient and effective power to the load while also avoiding power supply component failure. This requirement is based on the active and active energy flow rates in each location. The imbalance limitations specify that voltages near the server must stay within the lower and upper limits of the server's nominal voltage. The locations

and sizes of capacitor banks are limited by financial and physical constraints, such as the limited number of capacitor banks that can be mounted on buses to meet the system's requirements. The available power provided by the shunt capacitor can efficiently suppress the residual current required by the active load components. The entire power supply system will be released to KVAR, delivered originally by the power supply generator, if the active load of the distribution systems is cancelled the capacitor positioned in the middle of the active load. As a result, the generators' entire capacity is accessible to produce effective power.[9] The capacitor banks can be connected to the distribution system too far or too far before the inductive load station, providing loading relative to the active power, but the system will not receive the full benefits of voltage upgrades and loss reduction that can be provided by setting the appropriate capacitor.

The installation of shunt capacitors bank has a very high-cost requirements, so capacitors should be kept away to result in investments that are low input costs, higher energy savings and better voltage profile. The problem of proper capacitor placement is estimated in this paper using the minimal value, as well as the correct position and size of the capacitors to be included in the radial distribution system.

A permanent (fixed) network capacitor positioned far away from the network is (0.5–0.7) of the entire length of the server for optimal benefits in power development and reduction of losses in the central power supply. This local approach is best used as the "Sixth Commandment" because few rural areas distribute cargo in the same way. As a result, using a computer system to pick the right locations and sizes of capacitors in the system to minimize small electrical power, the following technique is best suited to discover a capacitor bank in the same loads as the unallocated radial distribution system. [9] By verifying the minimum capacitor size utilized in each phase (node) of the server and computing the overall power loss of the model, the computer system calculates the optimal location of the model capacitor. The computer then picks the feed phase with the lowest net loss and sets the subsequent capacitors in the same manner. Some nodes have a lower power factor, therefore adding more capacitor banks raises the voltage while increasing the loss. The increased total power loss in some circumstances of the suitable capacitor area to increase the minimum voltage will be illustrated in this article. [9]

2.3.1 Optimal Location and Sizing of Capacitor using ETAP:

The voltage magnitude, phase angle, active and reactive power, and line power flow associated with each network bus are the four values to be calculated in the power flow computation. It is expected that the ETAP software system has been thoroughly evaluated for graphical power systems and that the suitable capacitor has been determined using a genetic algorithm approachMany portions of the electrical system experience power outages as a result of naturally fluctuating loads and delivery equipment (transmission lines and transformers), resulting in reduced system capacity, power outages, and higher system losses. As a result, more powerful VARs are required in power systems. To install shunt capacitors in power systems, you must first decide the bank size in KVAR, and then you must select the position of the connection. It is an issue of efficiency to reduce the expenses associated with completing these two mathematical activities. As a result, development techniques should be employed. The ETAP module's Optimal Capacitor Placement (OCP) function is a sophisticated simulation tool that was created specifically for these applications. [10] Options for shunt capacitor placement in OCP systems to facilitate power supply and power enhancements while lowering total expenses. The accurate calculation technique automatically finds the proper position and size of the bank and allows you to evaluate the results with drawings and data on the number of capacitors in the buses, the voltage of each bus, the branch volume, and power loss statistics at the branches. The goal of using the right capacitor in ETAP is to lower the system's total cost, and in this case, to boost bus power by lowering power losses. To establish the acceptability of capacitor size and location, the cost of installation and operation is examined uniformly across all capacitor measurements. For capacitor placement, ETAP presently employs a genetic algorithm. Good qualities may be picked and handed on to the following generation with flexibility and crossover. [10] In the first repeated generations, the ultimate answer can be determined. Installation and running expenses, including maintenance, downtime, and cost reductions to decrease losses, may be expressed statistically by:

$$\text{O.F} = \text{Min}\left(\sum_{i=1}^{n} (XiC0i + Q1iC1i + BiC2iT) + C2 \sum_{i=1}^{N} T1PL \right)$$

2.4 FACTS Devices:

The fault current is defined as a rapid increase in power caused by a short circuit or o ther error in the power system .M Short circuits usually occur between rows or rows and ground in a three-phase transmission network. When a short-
circuit error, the current can increase up to 10 times the current load. This will have a negative impact on the reliability and efficiency of the power network. The distributi on network and transmission system make extensive use of the flexible AC transmiss ion system (FACTS) to help improve and control reliability and power consumption. The most used FACTS devices in the transmission network are Static VAR compens ator (SVC), thyristor-controlled phase shifting transformer (TCPST), thyristor-
controlled series capacitor or compensator (TCSC), STATCOM, SSSC, and UPFC. Additionally, STATCOM (DSTATCOM) distribution, dynamic voltage restorer (DV R), and unified power quality conditioner (UPQC) are used in the distribution networ k. The performance of FACTS devices is controlled by various controllers such as th e ambiguous controller, the dynamic controller, and the PI control. The complete pla cement of these FACTS devices provides high quality power upgrades. These FACT S devices can also be used to reduce the power available in the system. Researchers have considered first-
generation FACTS devices and reduced the level of power to reduce current errors. I n addition, assistive devices such as power electronic equipment-
based fault current limiter (FCL) and superconducting fault current limiter (SFCL) w ere used. However, now, without using the extra current limit, second-
generation FACTS devices can be used as the default current limits that keep the po wer level unchanged. By reducing errors during the cycle, these devices can significa ntly reduce the failure rate. The protection system can be reduced to stress. In this wa y, we can use a low-cost, highly cost-
effective protection system.[11] The FACTS system is used in this paper. Performanc e analysis of SSSC, STATCOM, and UPFC was performed as the current error reduc tion device.

2.5 Distributed Generation:

Distributed Generation is characterized distinctively by business sectors and organizations all around the globe, characterized either on the output yield they produce

or where they are arranged in the entire system. [13] The most well-known definition DG could fall into is a source of power arranged in the distribution system or customer side of the system. International Energy Agency (IEA) has characterized Distributed generation as a generation plant serving a consumer on location or offering help to a distribution network, associated with the network at distribution level voltages. [14] Electric Power Research Institute characterizes distributed generation as generation from a couple of kW up to an intensity of 50 MW. Then again, CIGRE characterizes DG as the generation, which has the attributes characterized as: It is not dispatched centrally at present; It is linked with the distribution network usually; It has a smaller size than 50–100 MW.

A huge advancement has been seen in the DG technology market in recent years, this is because of the greater demand in green energy and decentralization of power sources. There are several DG technologies present that can be found in the market today and research on many is still ongoing. Microturbines, combustion gas turbines, reciprocation engines, fuel cells, photovoltaic, and wind turbines all come under the category of distributed generation. Each of these technologies is responsible for providing a unique set of qualities and advantages and research is being conducted to get improved results. Diesel or gas reciprocating engines used to make most of the share of DG in the past but photovoltaic and more advanced wind turbines have taken up much of the market now while the reciprocating engines are being researched. [4]

Chapter 03: Practical Approach:

3.1Distributed Generation Model:

Distributed Generators (DGs) are power production systems whose output is not con nected to a central grid structure for long-
distance transmission to the point of use, but rather is located at the point of use, requ iring little or no transportation. Renewable energy technologies, such as solar photov oltaic and solar thermal systems, which harness solar energy from the sun and conver t it into usable forms of energy such as heat and electricity, are ideal candidates for d istributed energy generation due to their modularity and portability, as well as the abs ence of the need to transport energy resources (fuel) from point of production to poin t of use. Small hydropower is another type of renewable energy technology used in d istributed generation systems.

3.2 Overview:

The original model was submitted to load flow analysis, and data was collected prior to running a code for the genetic algorithm approach for determination of the optimal size of distributed generation. After determining the proper size and bus number for DG placement, we added the DG to the model and ran the load flow study again, and the results are shown below. Distributed generation (also known as on-site generation or decentralized generation) is the generation of electricity for on-site consumption rather than transmission via the electric grid from a large, centralized facility (such as a coal-fired power plant). On ETAP, the Sahiwal grid model is shown below. Local electricity transmission options must be supported as economic growth outpaces energy supply expansion in some areas of the country, while other regions suffer capacity constraints in providing power where and when it is needed.

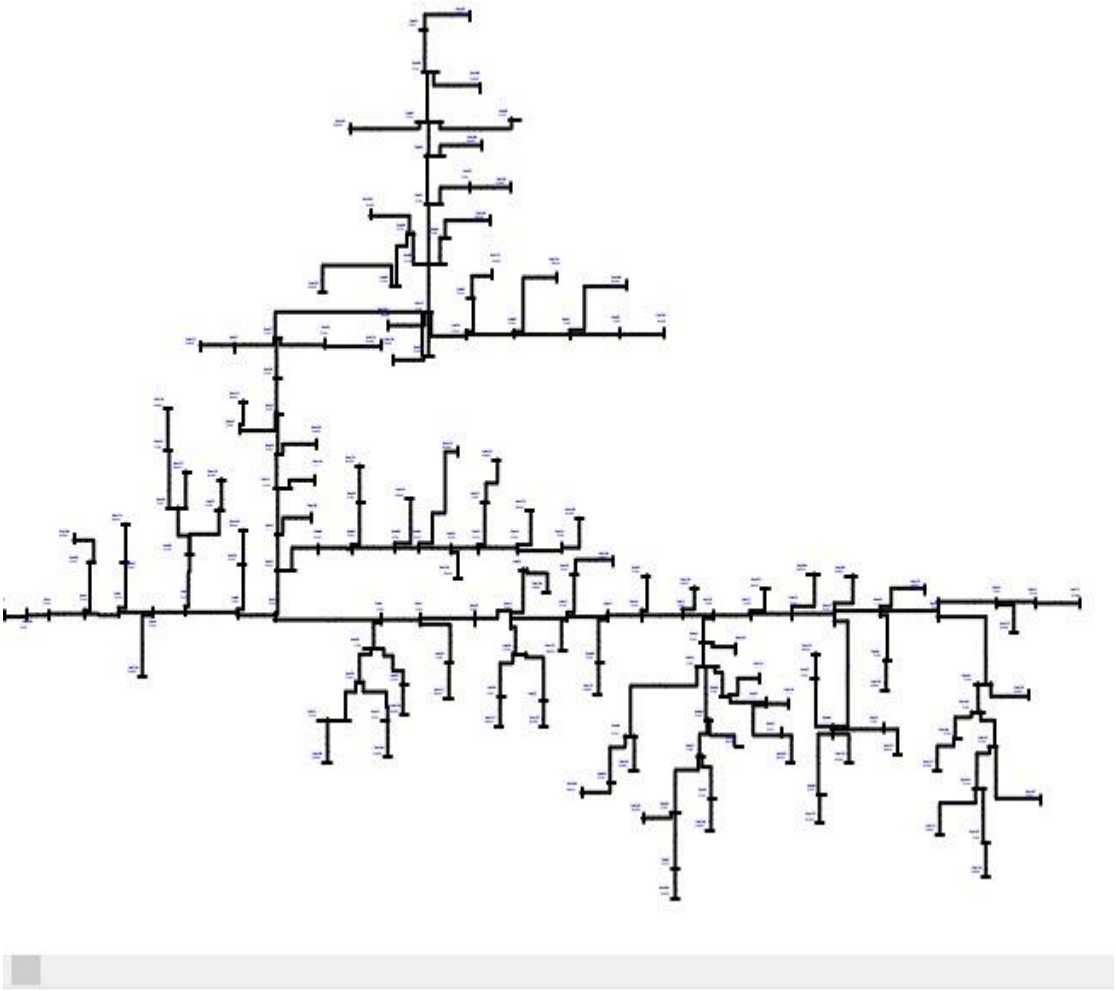

Figure 1 ETAP model of Sahiwal Grid of 177 busses.

3.3 Methods for installation of DG:

There are different methods for the allocation of dg for voltage profile improvement & loss minimization. The installation of the DG does require many other constraints to be taken care of as well. For example, when we talk about the DG optimum placement and location for Sahiwal Grid 177 Buses the working performance of all those DGs both of solar and wind type. Solar DG cannot work properly due to the angle adjustment as maximum energy harvesting. In that case the wind panels are not also the ideal ones to work due to the reason of not the presence of ideal wind case and wind speed Presence as well.

So, we have to find the optimum size and location by using the load flow method and then we can find the minimum losses position by positioning different size of distributed generator by using Genetic Algorithm technique.

3.3.1 Only Regulation is attained:

One approach is utilized to choose the best DG site just to optimize the voltage profile. This technique is divided into two stages. In the first step, voltage-sensitive buses (those with low voltage magnitude) are evaluated and prioritized, the DG is installed in all buses, and the voltage profile of the whole system in each installation is studied. After installing DG on each bus, all voltage profiles are ranked from best to worst. Finally, two lists are compared in order to determine the optimal location in the test system for the DG to be installed in order to create a decent voltage profile.

The computational way to implementing the first method is shown below.

- The load flow must be executed.
- The distribution system's voltage profile is determined following load flow.
- The priority list is based on load flow and the most sensitive buses (which should have voltage control).
- DG is then installed in each bus.
- After DG installation in each bus, the system load flow must be conducted.
- Another priority list is then created in this structure, and following comparison, they must be ranked from best to worst.
- The correct location of the DG is determined by comparing the priority list of Simulation Results: The load flow is performed on each distribution bus.

3.3.2 Both regulation & Losses reduction are Obtained:

Another method is used to establish the ideal DG location to optimize the voltage profile and reduce system losses. Voltage sensitive buses (those with low voltage magnitudes) are examined and prioritized in the first step. It is necessary to analyses the losses from least two biggest after installing DG in each bus. The ideal position for both loss reduction and voltage increase is therefore easily determined. The computational method for executing this technique is depicted below.

- Put out the data in MATLAB file all along with the Bus data and with line data as well, form a Y-bus matrix and then write a code of Newton rhopson method.
- It is necessary to run the base case load flow.
- Find out the line losses before the plantation of DG in MATLAB.

- By finding that We should then run a Genetic Algorithm technique code to find out the Value and position of the distributed generator at which we got the minimum losses and maximum voltage profile list.
- We may set number of DGs and limit of the power given by each DG to the system.
- We should set a priority list according to the given DG values and position. This priority list can also be defined by Voltage profile, total line losses and economic analysis as well.
- Then the DG in the best place is located and at last it is necessary to run the load flow program to reconfirm the validity of the line losses reduction and voltage profile improvement.
- We used the technique of genetic algorithm for finding out the optimum location and optimum value for DGs.

3.3.3 Genetic Algorithm:

Distributed Generation (DG) technology has continually improved in performance. DG sources include diesel, fuel cells, and renewable energy (RE) sources including photovoltaic (PV) systems and wind turbines. As a result, optimizing the size and placement of DG units in the distribution network aims to lower system losses, enhance system voltage profile, and increase system stability, dependability, and security. Researchers have focused their efforts on developing methods to reduce power losses. Analytical methods are only a couple of examples. Analytical calculations for appropriate DG distribution in order to minimize total actual power losses in DN are provided in the reference. The guide includes analytical approaches for DG allocation that minimize total actual power losses in DN.

Many strategies and solutions have been created that may be utilized to analyses DG allocation in a DN. The differences between these algorithms are in the description of the issue, the proposed approaches, and their assumptions. These reported methods include Genetic Algorithm (GA), Particles Warm Optimization (PSO), Fuzzy Logic (FL), Eigen-Value based Analysis (EVA), Artificial Neural Network (ANN), Ant Colony Optimization (ACO).

GA optimization is an effective method for identifying the optimal location of the DG in the distribution network. In general, the GA is run with a random population

set taken from the reachable solution space. The fitness function is evaluated for each solution, and the solutions that result are categorized. The population then develops through a variety of behaviors, including as reproduction, crossover, and mutation, to find the best fitness function and arrive at the ultimate optimal solution. The procedure is repeated until the termination condition is satisfied. until a criterion for termination is fulfilled. The main goal of the proposed technique is to find the best size and position for new DG resources by minimizing the system objective function. Two goals are utilized as a starting point for developing the formula in this study:

- **Reduction in losses**
- **Improving the Voltage Profile**

A flow chart of optimal distributed generation seating and scaling using GA.

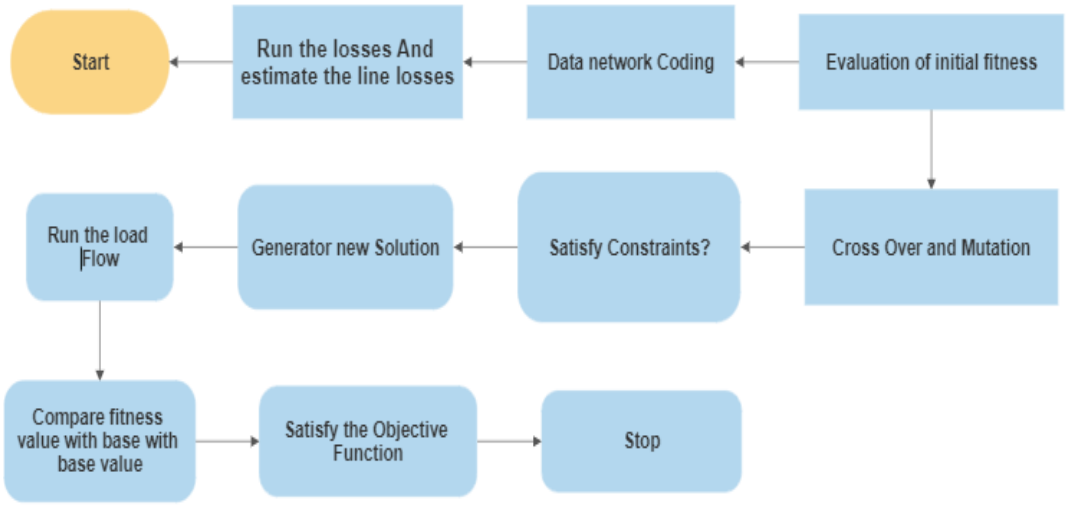

Figure 2: Genetic Algorithm

Genetic Algorithm results for the system gave us the Optimum position as well as the value of the Optimum DG location The results for the voltage profile and voltage regulation improvement can be seen after placing the particular DG in position in MATLAB and ETAP function. The graphs can explain the value.

Number of DGs	**2**
Location of DG 1	**12**
Location of DG 2	**48**
Value of DG 1	**1.8MW**

Value of DG 2	**1.1MW**

Table 1 Optimum Placement and value of DG

Sr. No	Network Type	Losses in MW	Losses in MVAR
1	Without DG and OCP	76.7	371.3
2	With OCP only	72.7	346.3
3	With DG (one Solar and one wind)	53.3	230.1
4	With DG (both Solar)	39.8	160.7
5	With both DG and OCP	35.7	135.5

Table 2 Losses with and without Rehabilitation

The graphs for improvement are shown below.

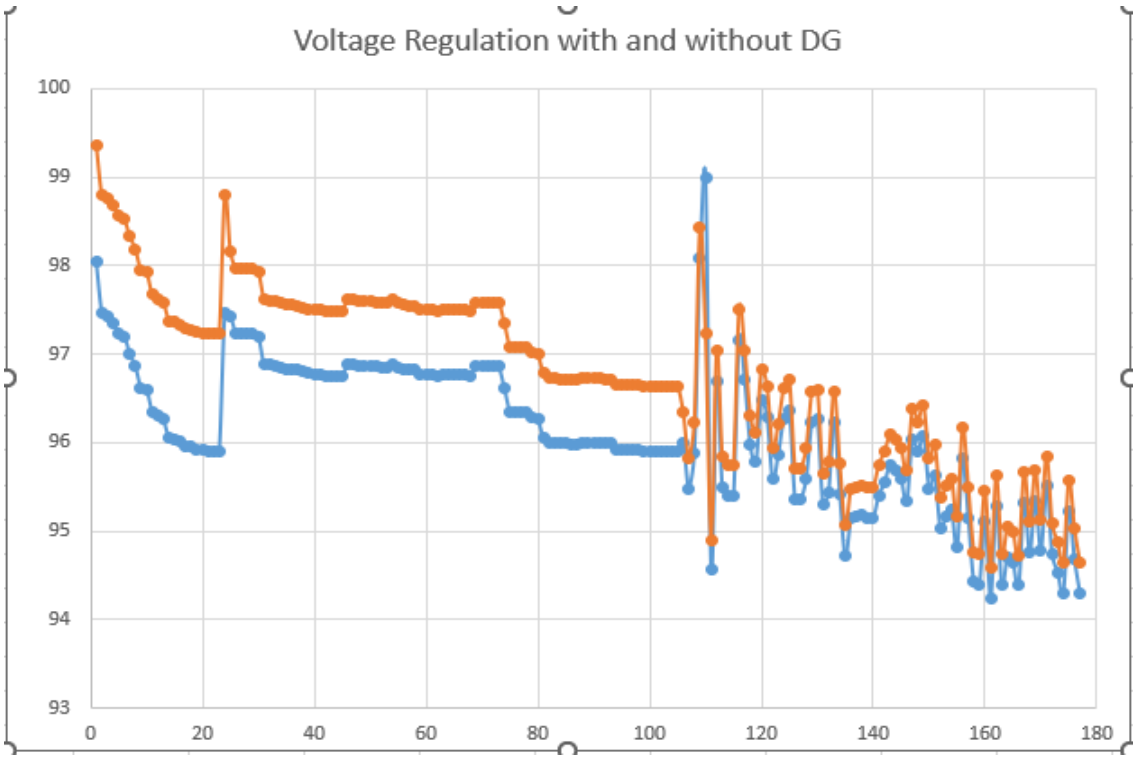

Figure 3: Voltage Regulation with and Without DG (one solar One wind)

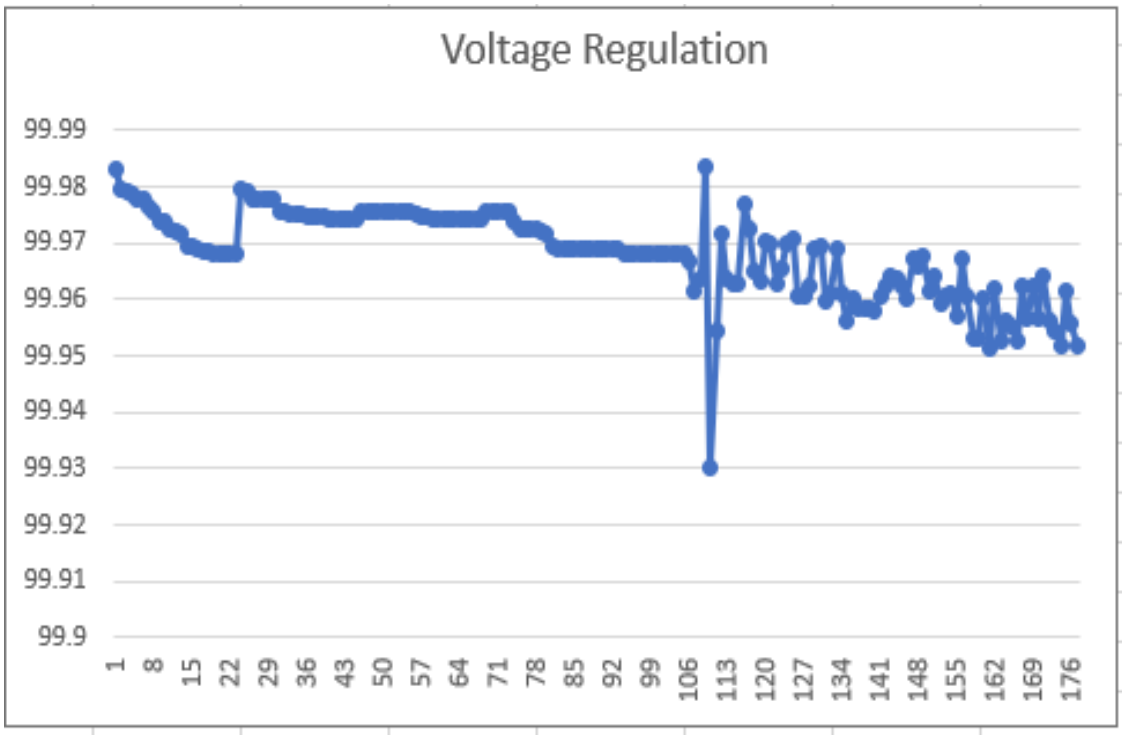

Figure 4: Voltage Regulation with and Without DG (one solar One wind)

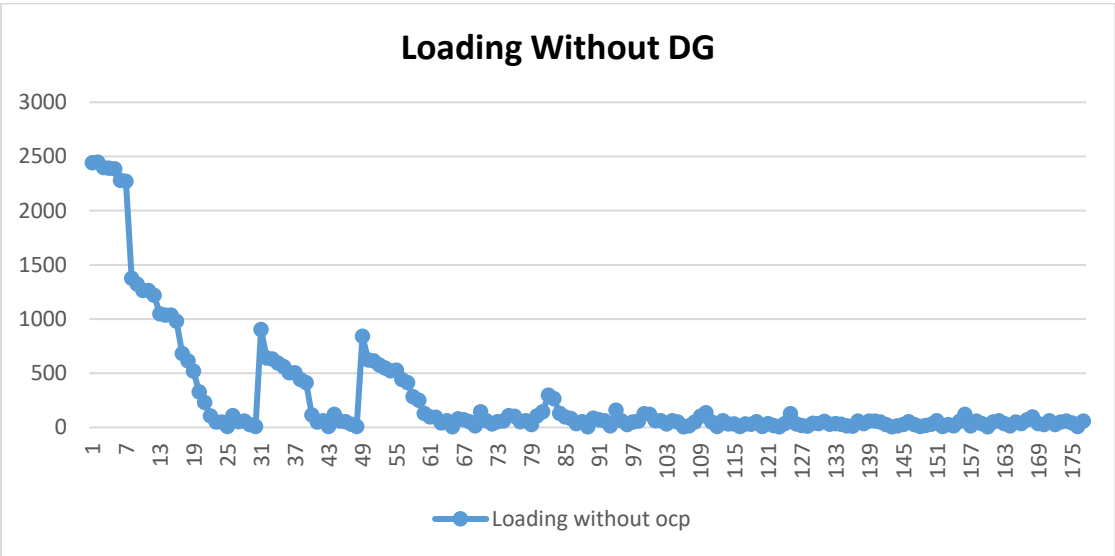

Figure 5: Loading without DG

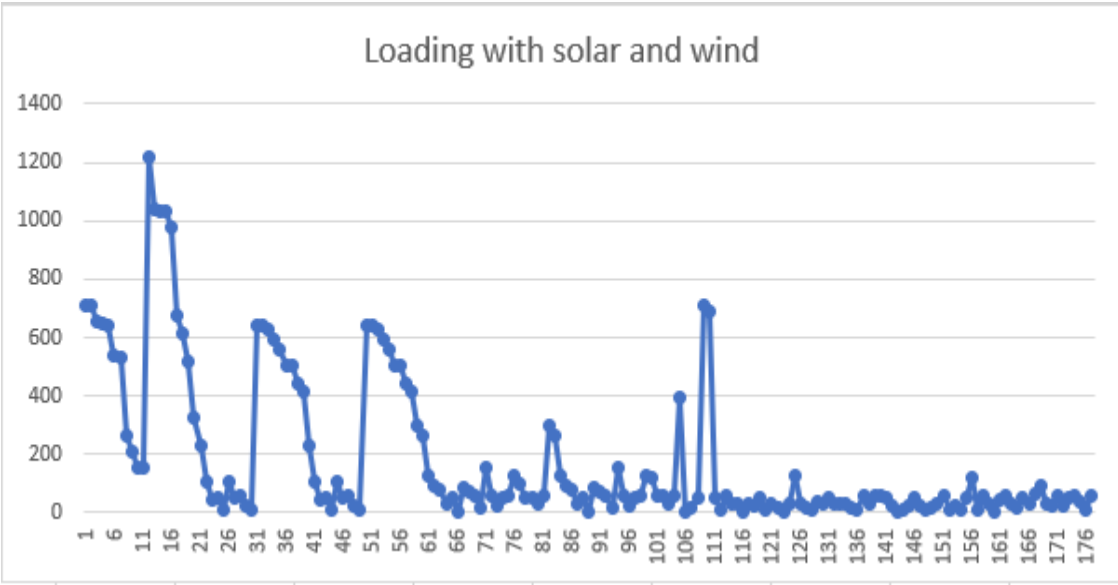

Figure:6 Loading with Solar and Wind

We used both Solar and Wind as our DG and find whether which one these are economically optimum and has best regulation impact over other. We also have to look at whether the plantation of either system is possible there or not.

We used both OCP and DG at same time and check its impact economically as well over voltage regulation of system. DG we used in Both OCP and DG is one wind and one solar case.

For the third case distributed generation and optimal capacitor placement were combined. The voltage regulation was improved profoundly but the cost outweighs the benefit in this case. The voltage profile are shown below.

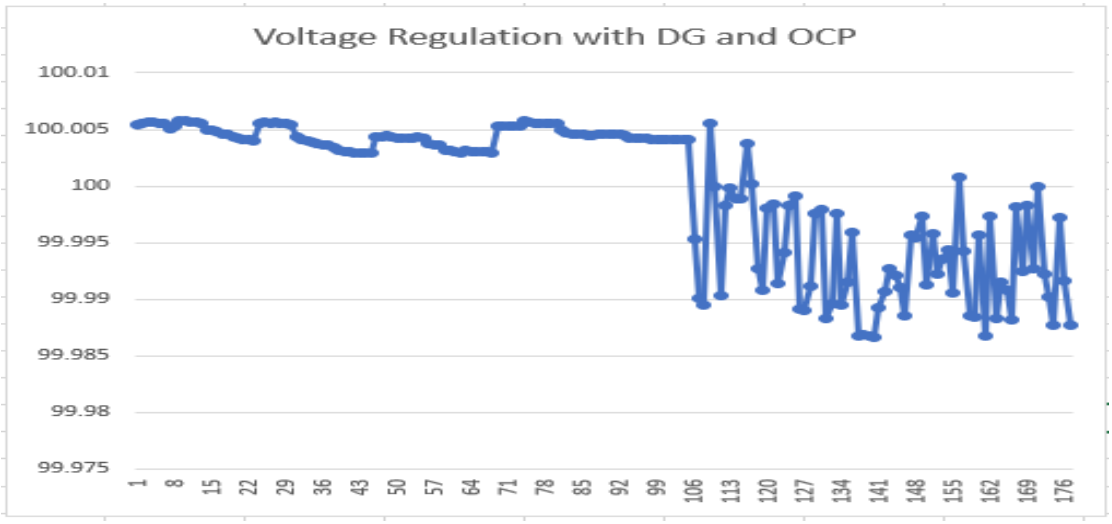

Figure 7:Voltage Regulation both OCP and DG (one solar One wind)

3.4 Installation of capacitor bank:

There are several methods for determining the size and position of capacitors.

- The most typical (intuitive) strategy relies on rules of thumb, with multiple load flow studies used to fine-tune the size and position. This method may not provide the best solution and is time-consuming and inefficient for large systems.

- The second way involves using the ETAP Optimal Power Flow (OPF) tool to optimize capacitor sizes depending on the engineer's candidate placements. Because OPF can optimize capacitor sizes but not locations, this technique requires per-selected locations.

- The Optimal Capacitor Placement (OCP) program which optimizes capacitor sizes and positions while taking cost into account, is the most effective way. OCP uses a genetic approach.

We used the third method which is Optimal Capacitor Placement (OCP). It is built in command in ETAP 19.0. We used this because it gives both optimal size of capacitor and location of capacitor for which maximum losses can be reduced. It is also important because it is economical. It allocates the minimum number of capacitors for which maximum losses can be reduced. This OCP tool works with a method in such a way that it connects the capacitor with every bus and changes its value from minimum to maximum and check for which location and size of capacitor the maximum reactive power is injected to the power system. The OCP tool note that value of bus number and the value of capacitor and connect optimal value capacitor to the optimal bus. Actually, due to inductive loads the reactive power taken by inductive loss has become increase due to which the capacitor banks inject the reactive power.

Number of Capacitors	05
Location of capacitor 01	04 BUS
Location of Capacitor 02	42 BUS

Location of Capacitor 03	10 BUS
Location of Capacitor 04	15 BUS
Location of Capacitor 05	18 BUS
Size of Capacitor 01	0.50 MVAR
Size of Capacitor 02	0.477 MVAR
Size of Capacitor 03	0.5 MVAR
Size of Capacitor 04	0.5 MVAR
Size of Capacitor 05	0.5 MVAR

Table 3: Optimum Placement and Value of Capacitors

We used the Optimal Capacitor Bank (OCP) tool for Sahiwal grid station. Optimal Capacitor Bank tool connected the capacitors with 5 buses with optimal values of capacitors. The size and location of capacitors is given below.

3.5 Results with OCP:

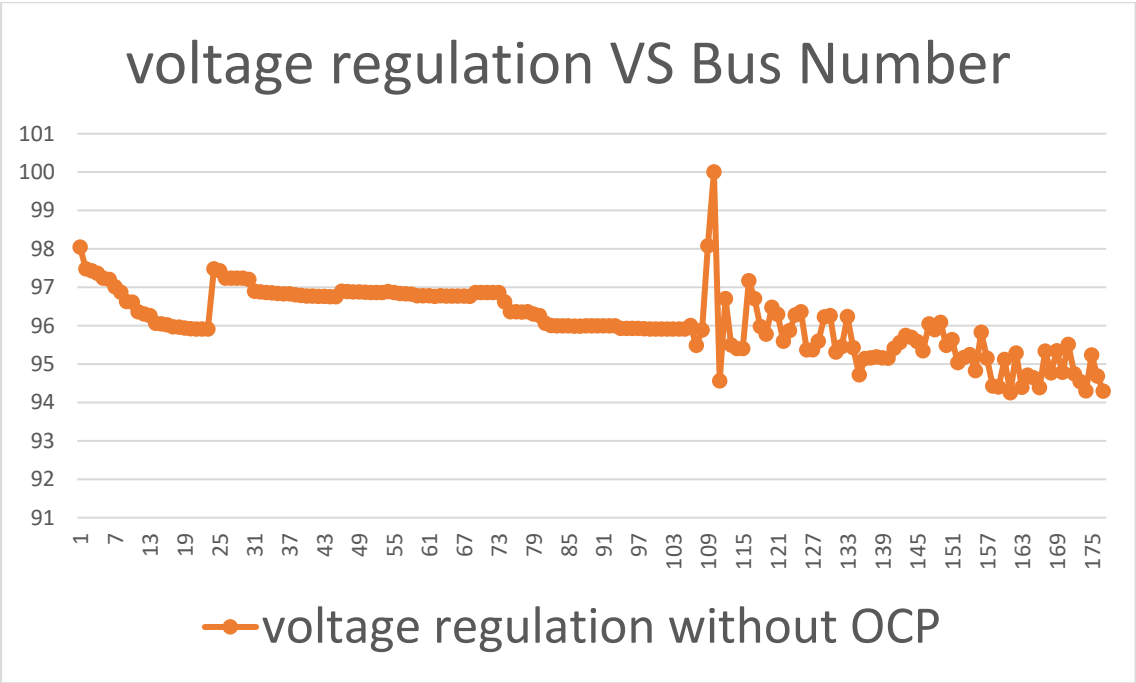

Figure 8: Voltage Regulation without OCP

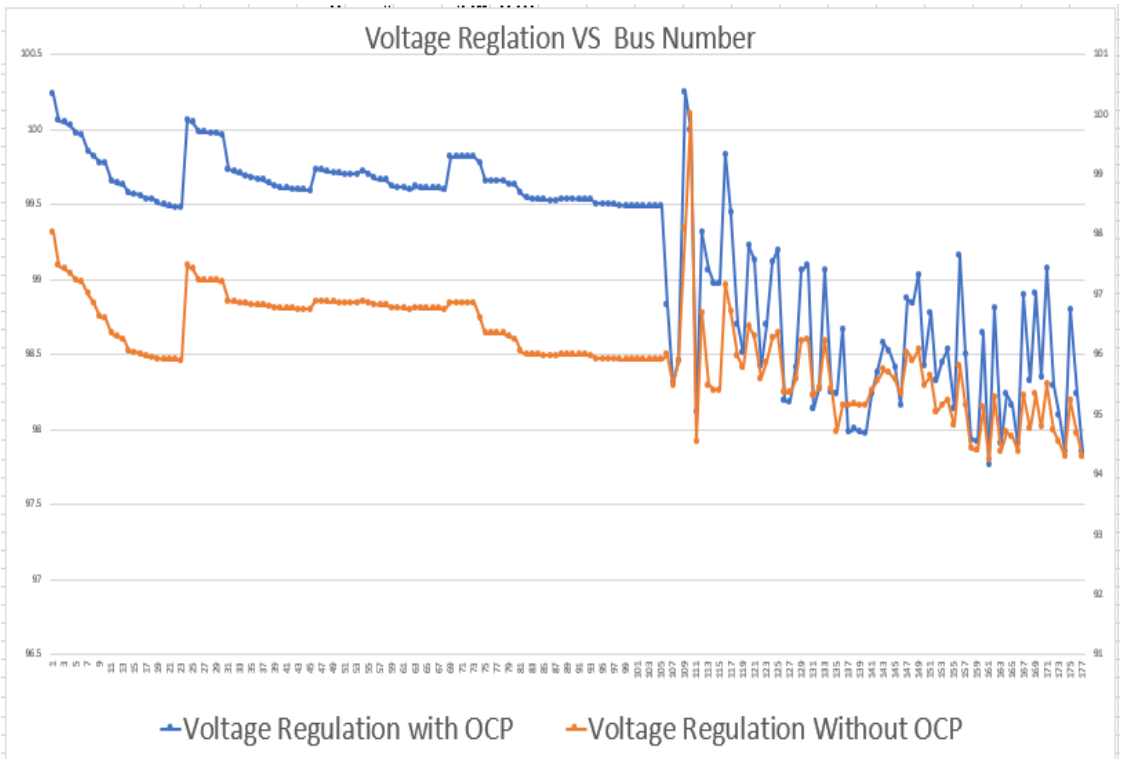

Figure 9: Voltage Regulation with and without OCP

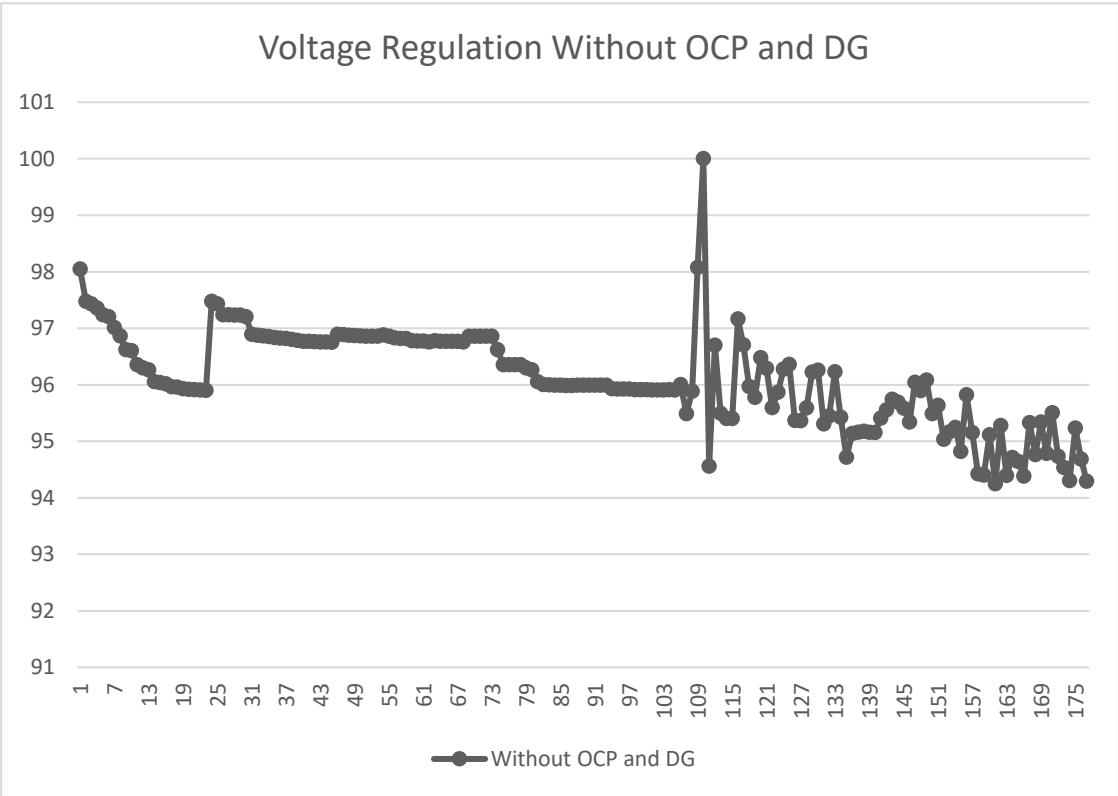

Figure 10: Voltage Regulation without OCP and DG

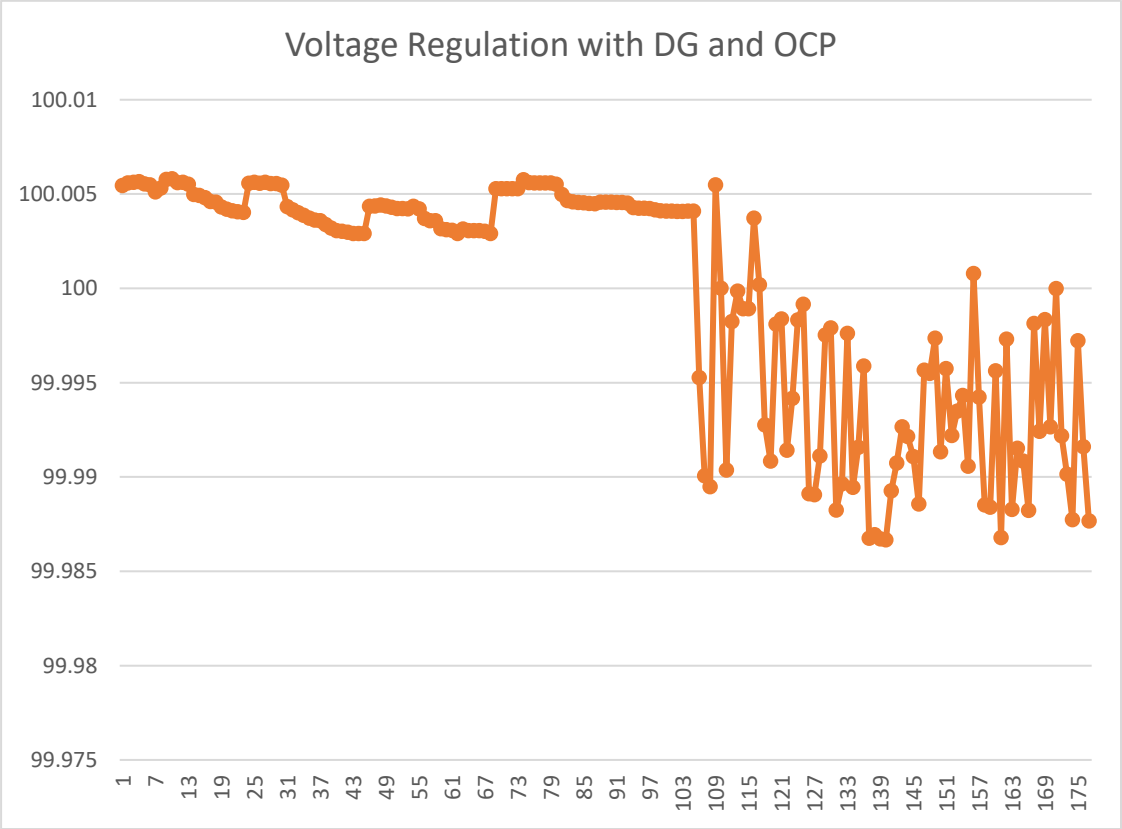

Figure 11: Voltage Regulation with OCP and DG

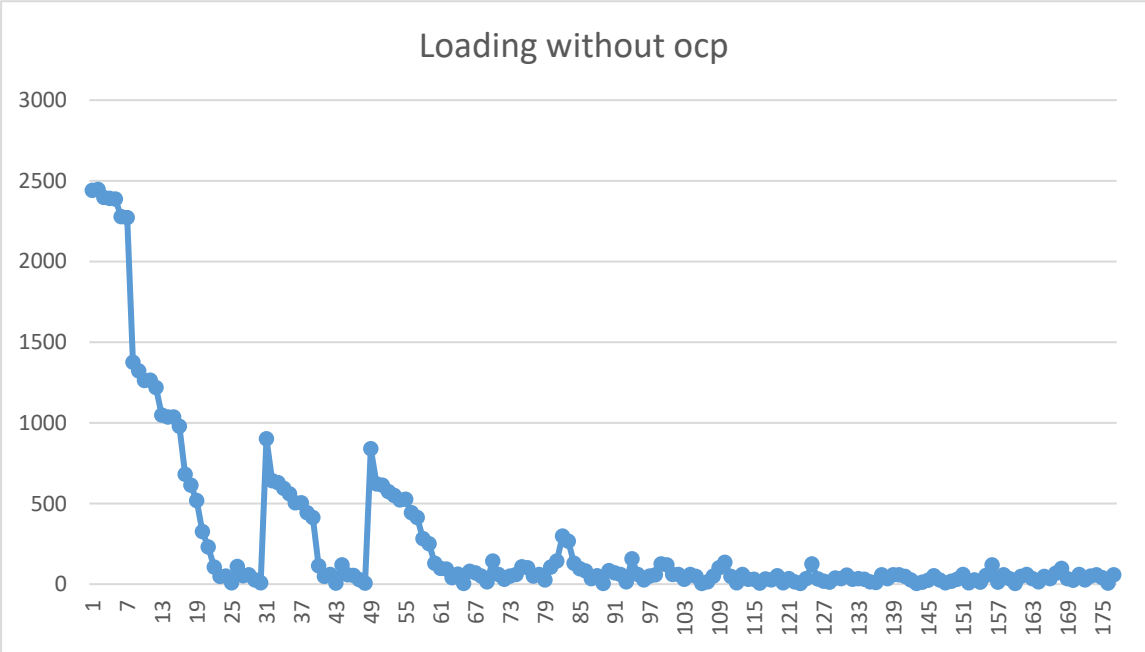

Figure 12: Loading without OCP

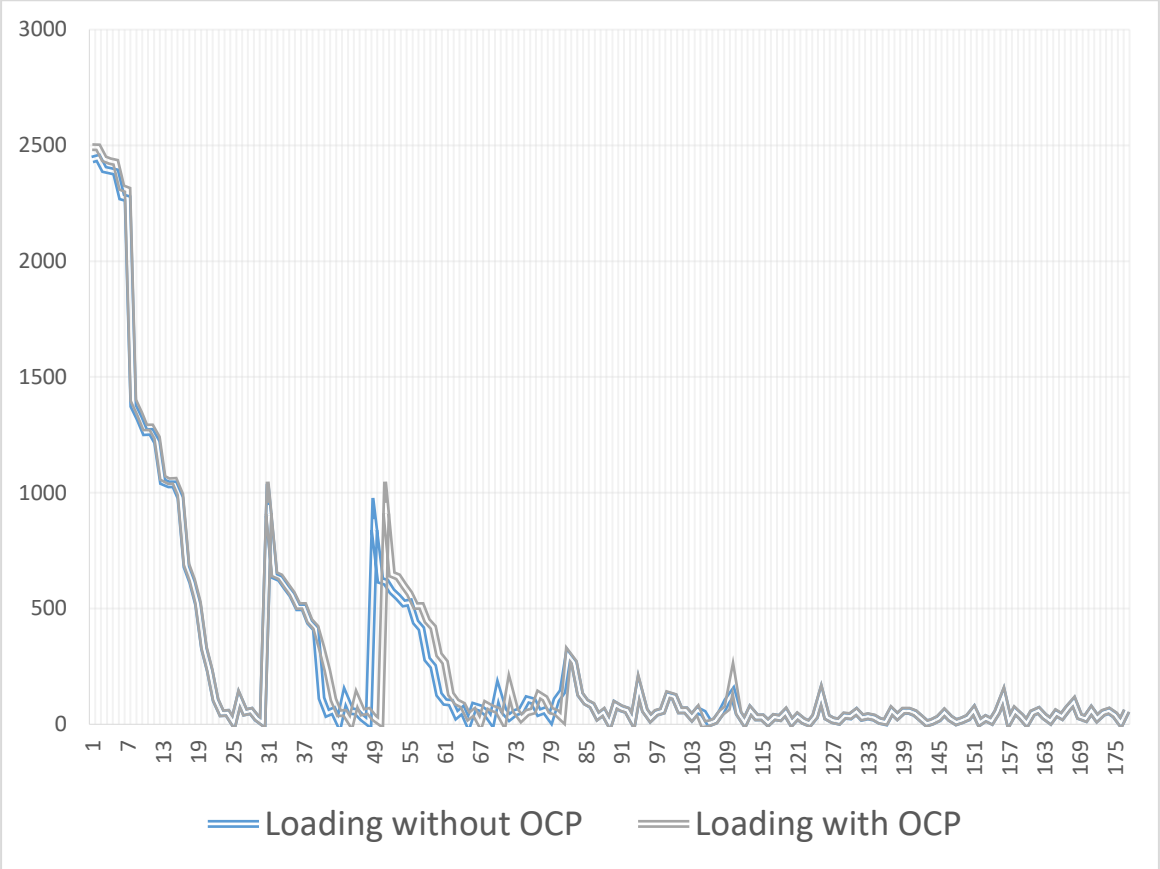

Figure 13: Loading with and without OCP

Chapter 04: Economic Aspect

4.1 Economic Analysis of OCP

$$One\ time\ Purchase\ cost\ of\ 5\ Capacitor\ Banks = 5000\ \$ \times 5$$

$$= 25{,}000\ \$$$

$$Maintainance\ Cost\ 4\%\ annualy = 0.04 \times 25{,}000$$

$$= 1{,}000\ \$$$

$$Planning\ Time = 10\ years$$

Using Eq:

$$Future\ Value = A(1 + \mathrm{i})^{n} \qquad 5.1$$

$$Future\ Value = 25{,}000(1 + 0.10)^{10}$$

$$= 64844\ \$$$

Using Eq:

$$Future\ Value\ of\ Annuity = \{A\ [(1 + \mathrm{i})^{n} - 1]\} / i \qquad 5.2$$

$$Future\ Value\ of\ Annuity = 1000\ \left(\frac{(1 + 0.1)^{10} - 1}{0.1}\right)$$

$$= 15937\ \$$$

$$Value\ at\ the\ end\ of\ investment = Future\ Value + Future\ Value\ of\ Annuity$$

$$= 80781\ \$$$

Losses without OCP = 76.6 kW

Losses after OCP = 71.6 kW

$Total\ Energy\ Saved$ = (76.1 – 71.6) kW

= 4.5kW

$Total\ units\ of\ Enery\ saved\ (kWhr)$ = 4.5kW × 365$days$ × 24$hours$

= 39420$kWhr$

$Per\ unit\ price\ of\ electricity\ in\ Pakistan$ = 23.9~24 $Rs.\ /kWhr$

$Total\ Cost\ of\ Energy\ Saved\ @\ Rs.\ ^{14}/kWhr$ = 5113.9 $

$Future\ Value\ of\ Annuity\ for\ cost\ of\ energy\ saved$ =

$$=5113.9\ \left(\frac{(1+0.1)^{10}-1}{0.1}\right)$$

= 81502.3 $

$Insight\ Into\ Investment$ = (81502.3 − 80781) $

= 721.3 $

As the difference is positive so the investment is justified.

Now we calculate the time in which the capacitor bank installation will complete its cost

$$((25{,}000/5113.9) \times 0.10) + 1 = (1 + 0.10)^n$$

$$\ln(1.488) = n \times \ln(1.10)$$

$$n = 4.16\ years$$

So, the capacitor Banks Installment will complete its cost in around 4 years.

4.2: Economic Analysis of Distributed Generation

$$Capital\ Cost\ of\ \text{wind}\ Generator = 2900^{\$}/kW \times 1800\ kW$$

$$= 5{,}220{,}000\ \$$$

$$Capital\ Cost\ of\ \text{solar module} = 1300^{\$}/kW \times 1150\ kW$$

$$=1495000\ \$$$

Case I: Using 2 solar units

Losses without DG (2 solar units) = 76.1kW

Losses after DG (2 solar units) = 44.3kW

For 2 units: 1495000 × 2 = 2990000 $

Losses reduced in this case = 76.1 – 44.3

= 31.8kW

$$Maintainance\ Cost = 3\%\ of\ Capital\ Cost\ annualy$$

$$= 89700\ \$\ annualy$$

$$Annual\ Saving = 31.8\ kW \times 24 \times 365 \times 14^{Rs.}/kWhr$$

3899952 / 185 = 21,080 $

Capital and Maintenance Cost:

Using Eq 5.1

$$Future\ Value = 2990000\ (1 + 0.10)^{10}$$

$$= 32{,}890{,}000\ \$$$

Using Eq 5.2

$$Future\ Value\ of\ Annuity = 89{,}700\ ((1+0.1)^{10}-1)\ /\ 0.1$$

$$= 1429586.9\ \$$$

Savings:

$$Future\ Value\ of\ Annuity = 21{,}080\ ((1+0.1)^{10}-1)\ /\ 0.1$$

$$= 335{,}960\ \$$$

$$Difference = 335{,}960 - (2990000+1429586.9)$$

$$= -4{,}083{,}626.9\ \$$$

As the difference is negative so it is not a good investment as it does not justify the installation of DG.

Case II: Using 1 solar and 1 wind unit

Losses without DG = 76.1kW

Losses after DG = 35.5kW

For 1 solar and 1 wind units: 1495000 + 5,220,000 = 6,715,000 $

Losses reduced in this case = 76.1 – 35.5

= 40.6kW

$$Maintainance\ Cost = 3\%\ of\ Capital\ Cost\ annualy$$

$$= 201{,}450\ \$\ annualy$$

$$Annual\ Saving = 40.6\ kW \times 24 \times 365 \times 14^{Rs.}/kWhr$$

4979184 / 185 = 26,914.5 $

Capital and Maintenance Cost:

Using Eq 5.1

$$Future\ Value = 6715000\ (1 + 0.10)^{10}$$

$$= 17{,}416{,}980\ \$$$

Using Eq 5.2

$$Future\ Value\ of\ Annuity = 201450\ ((1 + 0.1)^{10} - 1)\ /\ 0.1$$

$$= 3{,}210{,}594.2\ \$$$

Savings:

$$Future\ Value\ of\ Annuity = 26914.5\ ((1 + 0.1)^{10} - 1)\ /\ 0.1$$

$$= 428{,}947.8\ \$$$

$$Difference = 428947.8 - (6715000 + 3210594.2)$$

$$= -\ 9{,}496{,}646.4\ \$$$

As the difference is negative so it is not a good investment as it does not justify the installation of DG.

4.3: Economic Analysis of Distributed Generation and OCP

Losses without DG and OCP model = 76.1kW

Losses after DG and OCP model = 34.8kW

For OCP and DG (2 solar units): 25,000 + 2990000 = 3,015,000 $

Losses reduced in this case = 76.1 – 34.8

= 41.3kW

$$Maintainance\ Cost = 3\%\ of\ Capital\ Cost\ annualy$$

$$= 90{,}450\ \$\ annualy$$

$$Annual\ Saving = 41.3\ kW \times 24 \times 365 \times 14^{Rs.}/kWhr$$

5065032 / 185 = 27,378.5 $

Capital and Maintenance Cost:

Using Eq 5.1

$$Future\ Value = 3{,}015{,}000\ \ (1 + 0.10)^{10}$$

$$= 33{,}165{,}000\ \$$$

Using Eq 5.2

$$Future\ Value\ of\ Annuity = 90{,}450\ \ ((1 + 0.1)^{10} - 1)\ /\ 0.1$$

$$= 1{,}441{,}540.1\ \$$$

Savings:

$$Future\ Value\ of\ Annuity = 27{,}378.5\ ((1 + 0.1)^{10} - 1)\ /\ 0.1$$

$$= 436{,}342.8\ \$$$

$$Difference = 436{,}342.8 - (3{,}015{,}000 + 1{,}441{,}540.1)$$
$$= -\ 4{,}019{,}197.3\ \$$$

As there is no significant reduction in losses for this case hence it is not economically feasible. However, for this case, the voltage regulation is significantly enhanced as compared to the other two cases. This conclusion applies to this model depending upon the size of the grid station the results may variate.

Chapter 05: Power Quality Improvement

Voltage profile, Frequency profile, Harmonics and power supply reliability make up power quality. The degree to which the power supply approaches the ideal scenario of reliable, uninterrupted, zero distortion, and disturbance-free delivery is referred to as power quality.

Power quality concerns are garnering a lot of attention these days from both electrical utilities and end-users since they are costing industrial clients a lot of money. The most serious and commonly occurring PQ difficulties for sensitive loads are voltage sag and swell. PQ issues are defined as any variation or problem in voltage, frequency, phase angle, or current that causes a system to malfunction. The most common PQ issues are voltage sag and swell. These problems might be solved by designing equipment and electrical systems to minimize malfunctioning as a result of electrical disturbances. Investigate the medium that is conveying the electrical disturbance and try to eliminate or decrease its influence. Analyze the signs of a power quality issue to find out what's causing it and how to fix it. To remedy the symptoms of power quality concerns, use a power conditioning device and a specialized power device. Classification of power quality problem:

i) Voltage sag
ii) Harmonics
iii) Waveform distortion
iv) Frequency Problem
v) Electricity theft

5.1.1 Voltage sag:

A short circuit failure or an abrupt change in the characteristics of a power source or a load, such as during motor starting, can induce voltage sag. A voltage dip occurs when the voltage magnitude in the distribution or customer's electrical system drops for a brief period. Different failures in the transmission and distribution networks, faults in the linked equipment, or excessive inrush and switching currents in the customer's installation can all cause this.

Causes:

- Lightning

- Wind
- Trees that have fallen across power cables
- Digging into subterranean cables by construction personnel
- Squirrels and mice
- Equipment breakdowns
- Traffic accidents

Solution:

The voltage sag sensitivity of a machine is affected by its location. When the wiring is to blame, increasing wire size can help reduce voltage drop.

Sag correction devices, such as those found on the control panel, at the machine level or on the bus level, can typically be accessed through the plant service entrance. Uninterruptible power supply devices provide protection during voltage decreases, brief power shortages, and extended power outages.

5.1.2 Harmonics:

- The most popular representation of power quality is harmonic distortion, which displays the difference between the ideal sinusoidal waveform that the network voltage or load current should have and what occurs.
- Most power systems can take a certain amount of harmonic current, but problems develop when harmonics account for a significant portion of the overall load. As they travel through the power system, these higher frequency harmonic currents will have the following effects: Equipment failures as a result of high voltage distortion
- Circuit breaker nuisance tripping
- Generator Hunting
- Electronic Control Clock Errors
- Errors in metering necessitate true metering.
- Noise at the zero-crossing point
- Monitors that tremble and CRT screens
- High currents of load

Reduction of harmonics:

- In three-phase circuits with shared neutrals, separate neutral conductors are used. Additionally, each phase conductor is served by its own neutral conductor.
- Oversizing Transformers and Generators: As total current rises due to harmonic currents, transformer and generator thermal capacity should be increased. There is a consideration for oversizing.
- PF Control: Cutting-edge technology that removes harmonics introduced into the mains supply by nonlinear loads while still supplying inductive power required by the load, bringing the input power factor closer to unity. Active harmonics are used.
- Passive filters: Passive filters are used to tune a specific harmonic frequency. Harmonic currents flow in the filter rather than the supply because passive filters provide a low impedance channel for them.
- Special transformers have been created to handle the extra heating caused by harmonic currents.
- New true RMS meters can detect current up to kilohertz. These meters can be used to detect harmonic currents and provide accurate harmonic content readings.

5.1.3Waveform distortion:

Waveform distortions are common power issues that cause device failure and overloading of power sources. It's a sudden change in current and voltage waveforms as they travel through a device.

Any variation from the nominal sine waveform of the AC line voltage is referred to as voltage distortion. Current distortion has a similar description however voltage distortion and current distortion are not the same. Each has a different impact on loads and power systems, and each necessitates a different response.

Solution:

- The neutral conductor should be made larger. The neutral conductor path's impedance is reduced. The voltage drop and waveform distortion are reduced by lowering the neutral conductor impedance. Unfortunately, this type of wiring

change can only be made when the electrical distribution is installed for the first time.

- Lower the load on the branch-circuit panelboard. Every location has voltage waveform distortion. The ideas should increase your system's efficiency and lengthen the life of your equipment if it's the cause of equipment malfunction or overloaded power sources.
- Feeder circuits should not exceed 200 feet in length. This decreases the power system's overall impedance. Limit the length of branch circuits to less than 50 ft.
- Lighten the load on the transformer. You should not load a transformer to more than 60% of its kVA nameplate rating when it feeds electricity to nonlinear loads. This method prevents overheating and decreases distortion.
- Get rid of neutral conductors that are shared. It's preferable to run no more than five outlets off of one circuit breaker when wiring for electronic devices.

5.1.4 Frequency problems:

The divergence of the power system's fundamental frequency from its nominal value is referred to as power frequency fluctuations. A huge block of load is disconnected, or a large source of generation is taken offline due to failures in the bulk power transmission system. Frequency fluctuations are infrequent in modern integrated power systems.

The unexpected change in electrical loads on synchronous generators can cause the output power frequency to fluctuate substantially while the prime mover adjusts to match the demand.

The main reasons are uncertainty in wind and solar power generation, transactions in the distribution system's energy market and extreme load changes.

Solution:

- A secondary frequency response is an automatic 10% increase in a generator's output in reaction to a frequency drop that occurs within thirty seconds and can last up to thirty minutes.
- The high-frequency response is an indefinite decline in a generator's output in response to a frequency climb within ten seconds.

- A secondary frequency response is an automatic 10% increase in a generator's output in reaction to a frequency drop that occurs within thirty seconds and can last up to thirty minutes.

5.1.5 Electricity theft:

Mismanagement by the government, as well as power producing and distribution corporations, is blamed by citizens for the electricity shortages and load shedding.

Utility providers have been required to implement recovery-based load shedding, which implies greater load shedding in areas with low revenue recovery. Unfortunately, this has caused resentment in many communities because it is considered that customers who live in these regions yet pay their bills are being penalized as well.

The Sindh government, for example, has been chastised for failing to fund power distribution corporations sufficiently. They've also been accused of endorsing power-grabbers. Even though consuming energy without paying for it is unlawful and punishable by penalties or imprisonment, it is widespread and causes significant financial loss.

Individuals and communities steal electricity, but industries steal electricity as well by manipulating with meter readers. Theft is easier to commit when technology is outdated. Large businesses and industries can usually afford to pay their utility costs, so this is clearly an unwelcome problem.

Most inhabitants cannot afford the luxury of generators; therefore, power theft continues to be a problem throughout the country. Many people, especially those in low-income areas, suffer as a result of load shedding, which is used to manage restricted electric power generation and transmission. Many consumers no longer had the option of cooling off in air-conditioned shopping malls. Many were forced to endure up to 17 hours of load shedding in a single day through no fault of their own.

Power theft by a few individuals and communities affects schools, colleges, and small companies alike. Households in higher-income groups report less load shedding than those in lower-income areas, exacerbating the already widening economic gap.

The ever-increasing cost of electricity is one of the main excuses offered for power theft. We could argue that towns, neighborhoods, and people should hold each other to a higher level. We must, however, be aware of the notion that electricity is becoming a luxury item for the general public. There appears to be no light at the end of the tunnel for Pakistan unless we address the many challenges around transmission and distribution losses.

Chapter 06: Conclusion:

In this thesis, the rehabilitation of an 11kV Sahiwal grid 177 buses distribution network has been performed by using three rehabilitation techniques, namely, OCP, DG, and the combination of OCP and DG. Optimal capacitor placement has been carried out with the help of ETAP's Optimal Capacitor Placement Module while DG location has been determined by the help of Genetic Algorithm. All the three cases have exhibited significant enhancement in the voltage regulation of the system. however, only the first two cases have exhibited a significant reduction in real power losses of the DN. The economic analysis of the three rehabilitation techniques has been carried out by employing fundamental knowledge of economics regarding future value problems. The economic analysis presented shows that the first technique, i.e., OCP, is the most economically viable solution for the DN under consideration. However, the monetary values used for the economic analysis are only valid for the period in which the study was carried out and may not be applicable for the future analysis. Moreover, the thesis also discusses different methods to improve the power system quality, which includes the Removal of harmonics from DN system and also discuss the types of different filters used in these to prevent the harmonics. Moreover, we also discuss the methods for waveform distortion removal and their different methods along with the discussion of frequency profile improvement. We also have brief analysis over the utilities of all these over the system requirement as well.

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